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AUGUST/RESIT EXAMINATIONS 2024/2025

MODULE: CSC1048 - Computability and Complexity

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECGTE	Engineering & Computing GTE
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Progrgam EC

YEAR OF STUDY: 1,3

EXAMINERS: Dr. David Sinclair (Internal) (Ext. 5510)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer 4 Questions.
All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

Requirements for this paper (Please mark (X) as appropriate)

☐	*Log Tables*	☐	*Thermodynamic Tables*
☐	*Graph Paper*	☐	*Actuarial Tables*
☐	*Dictionaries*	☐	*MCQ Only - Do not publish*
☐	*Statistical Tables*	☐	*Attached Answer Sheet*

Section A

QUESTION 1 ***[Total marks: 25]***

1(a) [5 Marks]

What are function types in Haskell? How is a function type written in Haskell? What is the implication of the particular form Haskell uses for defining a function type and invoking functions?

1(b) [8 Marks]

What are *class constraints* in Haskell? Give an example of how to use a *class constraint*.

1(c) [12 Marks]

Write a Haskell function, $\mathbf{without\ using\ the\ reverse\ function}$, that tests whether or not a list is a "palindrome", i.e. the list is equal to its reverse. The function should use class constraints to ensure that it will work for lists of integers, floats, and characters.

[End Question 1]

QUESTION 2 ***[Total marks: 25]***

2(a) [5 Marks]

In the Haskell Programming language describe how *guards* operate with the use of example code.

2(b) [10 Marks]

Let a polynomial be represented as a list of coefficients. For example $a_3x^3 + a_2x^2 + a_1x^1 + a_0$ is represented as the list [a_0,a_1,a_2,a_3] . Define a Haskell function `evalPoly`, with the appropriate function type, which, given a polynomial and a value for x, will calculate the value of the polynomial for that value of x.

Hint: You may find the following identity helpful:
$a_nx^n + \ldots + a_2x^2 + a_1x + a_0 = a_0 + x(a_1 + x(a_2 + x(\ldots a_n)\ldots))$

2(c) [10 Marks]

Write a Haskell function `shortest`, with appropriate function type, that takes a list of lists and returns the shortest list in the list (and returns [ ] if the list of lists is empty).

[End Question 2]

Section B

QUESTION 3 ***[Total marks: 25]***

3(a) [5 Marks]

In *computability* what does it mean when a problem is *undecidable*?

3(b) [13 Marks]

Prove that the *Halting Problem* is undecidable.

3(c) [7 Marks]

Describe how *reducibility* can be used to show a problem is undecidable.

[End Question 3]

QUESTION 4 ***[Total marks: 25]***

4(a) [10 Marks]

Formally define a Turing Machine.

4(b) [15 Marks]

Outline the structure of the proof of the Cook-Levin theorem. In particular, discuss the number of clauses generated in the proof.

[End Question 4]

QUESTION 5 ***[Total marks: 25]***

5(a) [5 Marks]

Formally define the complexity class $\mathbf{P}$.

5(b) [5 Marks]

Formally define the complexity class $\mathbf{NP}$.

5(c) [15 Marks]

The 3-Colourability problem can be stated as “does there exist an undirected graph G whose nodes are coloured red, green, and blue such that no adjacent nodes have the same colour”?

Prove that 3-Colourability belongs to the class $\mathbf{NP}$ if Satisfiability belongs to the class $\mathbf{NP}$.

[End Question 5]

[END OF EXAM]